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Letter

Codeposition Strategy Enables Stable Built-In Electric Field in Sn−Pb Perovskite Solar Cells

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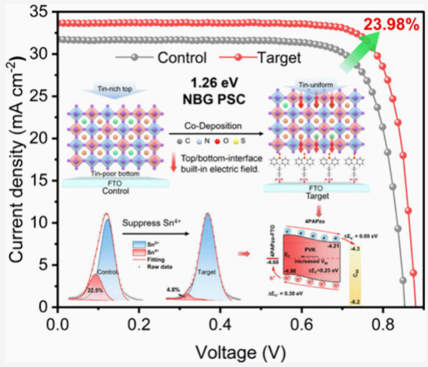
ABSTRACT: Tin−lead (Sn−Pb) perovskite is crucial for all-perovskite tandem solar cells, but its performance is limited by flattened built-in electric fields. Two factors cause this: fast SnI_2 crystallization, causing Sn accumulation at the top and deficiency at the bottom, weakening the bottom electric field; and acidic PEDOT:PSS oxidizing Sn^{2+} to Sn^{4+}, inducing p-doping at the top, further flattening the field. Here, a new self-assembled molecule, (4-(5,5-dioxido-10H-phenothiazin-10-yl)butyl)phosphonic acid (4PAPzo), was introduced into the perovskite precursor for codeposition. Its sulfonyl group coordinates with Sn^{2+} to suppress oxidation and guide more Sn to the bottom during film deposition. These synergistic effects establish stable and enhanced built-in electric fields at both interfaces without PEDOT:PSS. The resulting Sn−Pb device achieved a power conversion efficiency (PCE) of 23.98% and retained 88% of its initial value after 900 h of maximum power point tracking in nitrogen. The four-terminal all-perovskite tandem device achieved a PCE of 29.20%.



Metal halide perovskite solar cells (PSCs) are promising candidates for photovoltaics due to their excellent properties, such as high absorption coefficient[1] and tunable bandgap.[2] With a narrow-bandgap (NBG) in the range of 1.2 to 1.3 eV, mixed Sn−Pb PSCs not only theoretically achieve higher efficiency of single-junction cells,[3] but also serve as the bottom subcells in all-perovskite tandem solar cells (TSCs) to exceed the Shockley-Queisser limit of single-junction cells,[4] which represents a significant application potential.[5−8] However, the PCE of Sn−Pb PSCs is currently limited by significant open-circuit voltage loss (V_{OC}). This is mainly due to the weakening and disruption of the built-in electric field in the perovskite layer, which leads to a reduced efficiency in charge separation and transport. The root cause lies in the faster crystallization dynamics of Sn^{2+} cause it to accumulate at the top interface of the perovskite,[9−11] resulting in a Sn-poor region at the bottom (i.e., bottom Sn-poor perovskite). The uneven Sn/Pb vertical distribution causes the energy levels at the bottom interface to shift downward, while the Sn component accumulated at the top interface is more prone to oxidation, leading to p-doping at the top interfac.[9−11] Together, these factors promote the flattening of the energy band structure at both top and bottom interfaces, resulting in the reduced built-in electric field at the interfaces and increased V_{OC} deficit.

In recent years, constructing a strong built-in electric field at the interfaces in perovskite solar cells has been recognized as a key strategy for improving the V_{OC} and promoting the separation and transport of photogenerated charge carriers. To address the issue of weakened built-in electric fields caused by the lack of Sn component at the bottom in Sn−Pb devices, researchers have proposed various strategies. Liu et al., for example, effectively increased the Sn component at the bottom of the perovskite layer by predepositing a layer of SnI_2 on PEDOT:PSS, which enhanced the built-in electric field, promoted charge carrier separation and directional extraction, and achieved a PCE of 23.2%.[12] Wang et al. increased the distribution of Sn component at the bottom interface of the perovskite by incorporating oligomeric proanthocyanidins, which exhibit stronger chelation with Sn^{2+}, into PEDOT:PSS.[13] This strategy elevated the valence band maximum (VBM) at the bottom interface, induced band bending, and significantly improved the device's V_{OC}. Liu et al. increased the distribution of Sn component at the bottom by adjusting the annealing temperature, and effectively induced band bending in the inorganic Sn−Pb perovskite photodetector without hole-transport-layer (HTL).[14] Liu et al. suppressed the upward segregation of Sn by introducing indium ion (In^{3+}) into the precursor, achieving a longitudinally uniform distribution of

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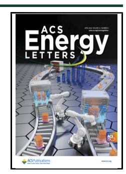

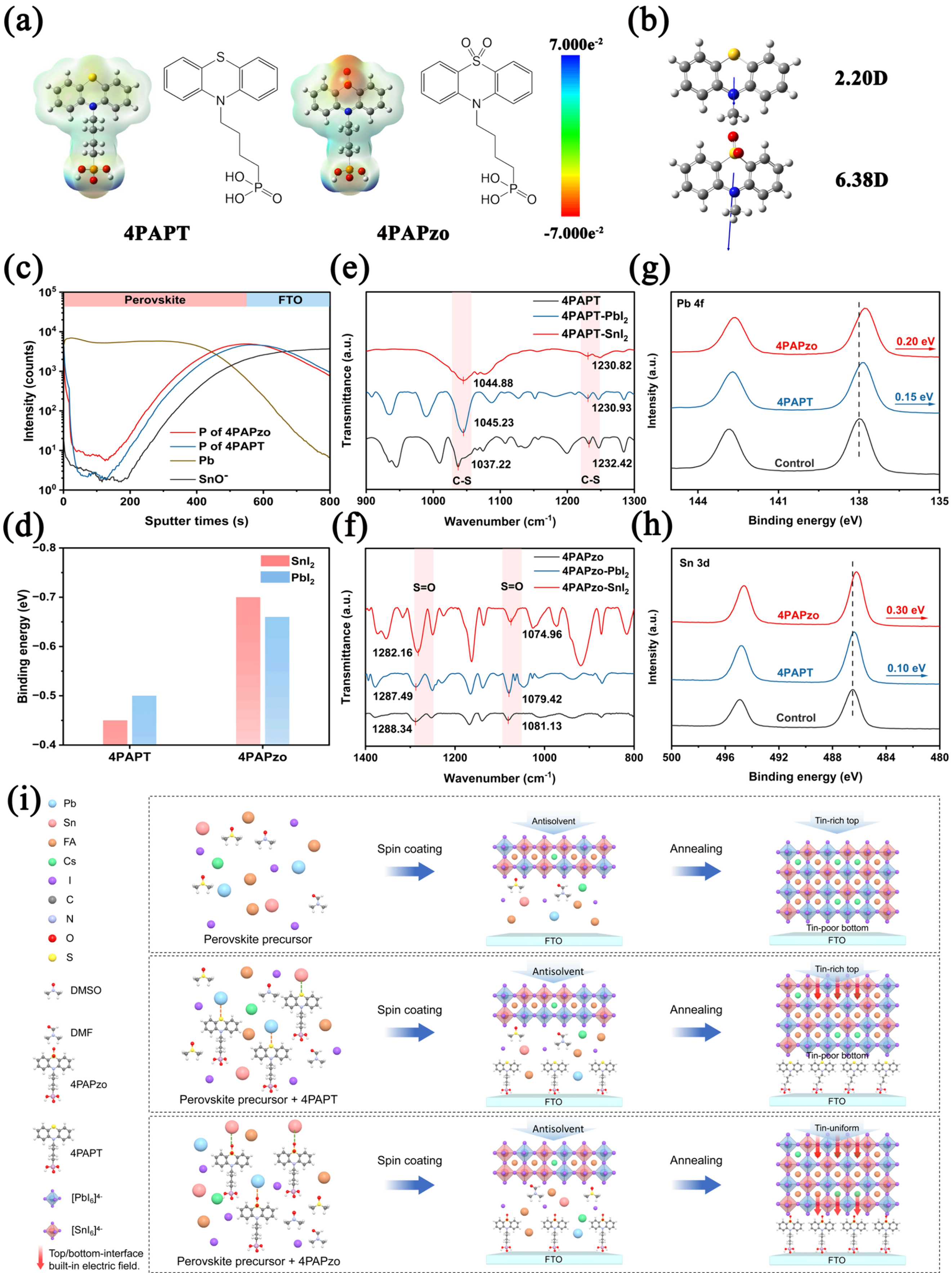

Figure 1. Electrostatic potential surfaces (ESP) and chemical structurals (a) and dipole moment (b) of 4PAPT and 4PAPzo. (c) ToF-SIMS depth profiles of the SAMs in films. (d) The binding energy between SAMs and PbI_2/SnI_2. (e) C−S stretching vibration of 4PAPT and its adducts with

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Figure 1. continued

PbI_2/SnI_2. (f) S=O stretching vibration of 4PAPzo and its adducts with PbI_2/SnI_2. XPS spectra of (g) Pb 4f and (h) Sn 3d of the perovskite film. (i) Deposition schematic of perovskite for Control, 4PAPT-doped, and 4PAPzo-doped devices.

Sn/Pb components.[11] These results show that a stable built-in electric field at the bottom interface requires a uniform vertical distribution of the Sn component in Sn−Pb devices. Therefore, to further minimize the voltage loss in Sn−Pb devices, constructing a stable and enlarged built-in electric field at bottom interface via unifying the Sn distribution becomes essential.

Another key issue that is often overlooked is the oxidation of Sn^{2+} by PEDOT:PSS and its impact on the built-in electric field. The commonly used HTL, Poly(3,4-ethylenedioxythiophene):polystyrenesulfonate (PEDOT:PSS), is hydrophilic and strongly acidic,[15,16] which accelerates the oxidation of Sn^{2+} in the perovskite layer,[17] forms Sn vacancies, and generates defects at the interface and throughout the perovskite layer.[18] While Sn components are enriched at the top interface, the oxidation of Sn^{2+} and corresponding p-doping effect ultimately leads to a degradation of the top-interface built-in electric field, which also impairs the long-term performance of the device. To replace PEDOT:PSS, poly[bis-(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) was also used. However, its hydrophobicity poses significant challenges for large-area coating of the perovskite layer, reducing commercialization potential.[19,20] Another inorganic HTL, nickel oxide (NiO_x), was also demonstrated. However, it contains Ni^{3+}, which also oxidizes Sn^{2+}.[21,22] On the other hand, although various modification strategies have been developed for commonly used PEDOT:PSS, such as additives[23,24] and interfacial modification,[25,26] these methods usually increase process complexity and have not fundamentally solved the long-term stability problems of PEDOT:PSS. Therefore, it is urgent to address HTL-related stability issues and prevent p-doping at the top interface to maintain a stable top-interface built-in electric field in Sn−Pb PSCs.

Notably, self-assembled materials (SAMs) have demonstrated excellent hole extraction efficiency when used as HTLs in Pb-based PSCs, which has led to significant breakthroughs in the efficiency of Pb-based PSCs in recent years.[27−30] When SAM molecules are introduced into the perovskite precursor to enable the codeposition of the HTL and perovskite layer, SAMs not only spontaneously migrate to the bottom interface via their phosphate groups to function as hole transporters but also interact with components in the perovskite precursor solution to regulate its crystallization. Tian et al. for example, introduced the SAM material (4-(2,7-dibromo-9,9-dimet hylacridin-10(9H)-yl)butyl)phosphonic acid (DMAcPA) into the perovskite precursor, where DMAcPA formed a chelate complex with PbI_2 and deposited at the bottom interface of the film. This reduced the PbI_2 content on the surface of the perovskite film and modulated the work function (W_F) of ITO.[31] In our previous work, (aminomethyl)phosphonic acid (AMP) was introduced into the perovskite precursor, which forms strong hydrogen bonds with FA^+ and sinks to the buried interface, thus inducing preferential crystallization of FA^+ and preventing Cs^+ aggregation at the buried interface.[32] So far, there have been no studies on the codeposition of SAM and perovskite precursors in Sn−Pb perovskite system, primarily due to the complexity arising from the different crystallization kinetics of Sn^{2+} and Pb^{2+}[33,34] and the susceptibility of Sn^{2+} to oxidation.[6] By developing a SAM with special functional groups for codeposition, it is possible to balance the crystallization rates of Sn^{2+} and Pb^{2+}, achieve a uniform vertical distribution of Sn/Pb, suppress the p-doping effect of Sn^{4+} in perovskite body, and form a stable top-interface and bottom-interface built-in electric field to synergistically improve the efficiency and long-term stability of devices.

Here, we introduce a novel SAM, (4-(5,5-Dioxido-10H-phenothiazin-10-yl) butyl)phosphonic acid (4PAPzo), which contains a sulfonyl group (O=S=O), and co-deposit it with the Sn−Pb perovskite precursor to construct a PEDOT:PSS-free Sn−Pb PSC with a stable built-in electric field. Due to the strong Lewis basicity of oxygen and the stronger Lewis acidity of Sn^{2+} compared to Pb^{2+}, the sulfonyl (O=S=O) group in 4PAPzo preferentially coordinates with Sn^{2+} according to the Hard and Soft Acids and Bases (HSAB) theory.[35] During crystallization, 4PAPzo directs Sn^{2+} to deposit at the bottom through its phosphate (PO_3H_2) structure and the anchoring effect of the hydroxyl group on the substrate.[36] This regulation alleviates the problem of bottom-interface built-in electric field weakening caused by excessive Sn accumulation at the top interface and Sn deficiency at the bottom interface. As the Sn content increases at the bottom, the valence band maximum (VBM) at the bottom of the perovskite layer is elevated, enhancing the bottom-interface built-in electric field. In addition, removing PEDOT:PSS fundamentally reduces the risk of Sn^{2+} oxidation, and 4PAPzo further mitigates p-doping induced by Sn^{2+} oxidation through the strong interaction between its Lewis basic oxygen and Sn^{2+}, thus enhancing the built-in electric field at the top interface. Benefiting from this synergistic mechanism, the quasi-Fermi level splitting (QFLS) of the PEDOT:PSS-free $Cs_{0.25}FA_{0.75}Sn_{0.5}Pb_{0.5}I_3$ PSC fabricated using the codeposition strategy reached 0.906 V and established a stable dark-state built-in electric field of 0.25 eV, achieving a remarkable PCE of 23.98% with excellent stability. It retained 88% of its initial efficiency after 900 h of maximum power point tracking (MPPT) under simulated one-sun illumination in nitrogen at 25 °C. In addition, the 4-T all-perovskite tandem device, constructed using the optimized NBG PSC and wide-bandgap (WBG) semitransparent PSC, achieved a PCE of 29.20%.

The electrostatic surface potential (ESP) of [4-(10H-phenothiazine-10-yl)buthyl]phosphonic acid (4PAPT) and 4PAPzo were calculated using density functional theory (DFT), as shown in Figure 1a. 4PAPzo was designed by introducing a sulfonyl (O=S=O) functional group into 4PACz, while 4PAPT was designed by incorporating a sulfur (S) functional group. The chemical structures of 4PAPzo and 4PAPT were confirmed by nuclear magnetic resonance (^1H NMR and ^{13}C NMR) spectra, as shown in Figure S1−S4. The electron density on the sulfonyl group in 4PAPzo is much higher than that on the S atom in 4PAPT. This suggests that the O atoms in 4PAPzo have a stronger coordination ability with Sn^{2+} and Pb^{2+}, enabling more effective passivation of defects on the perovskite surface and grain boundaries. The dipole moments of 4PAPzo and 4PAPT are 6.38 and 2.20 D, respectively (Figure 1b), while that of 4PACz is 1.84 D (Figure S5). The dipole moment of 4PAPzo is much larger than those

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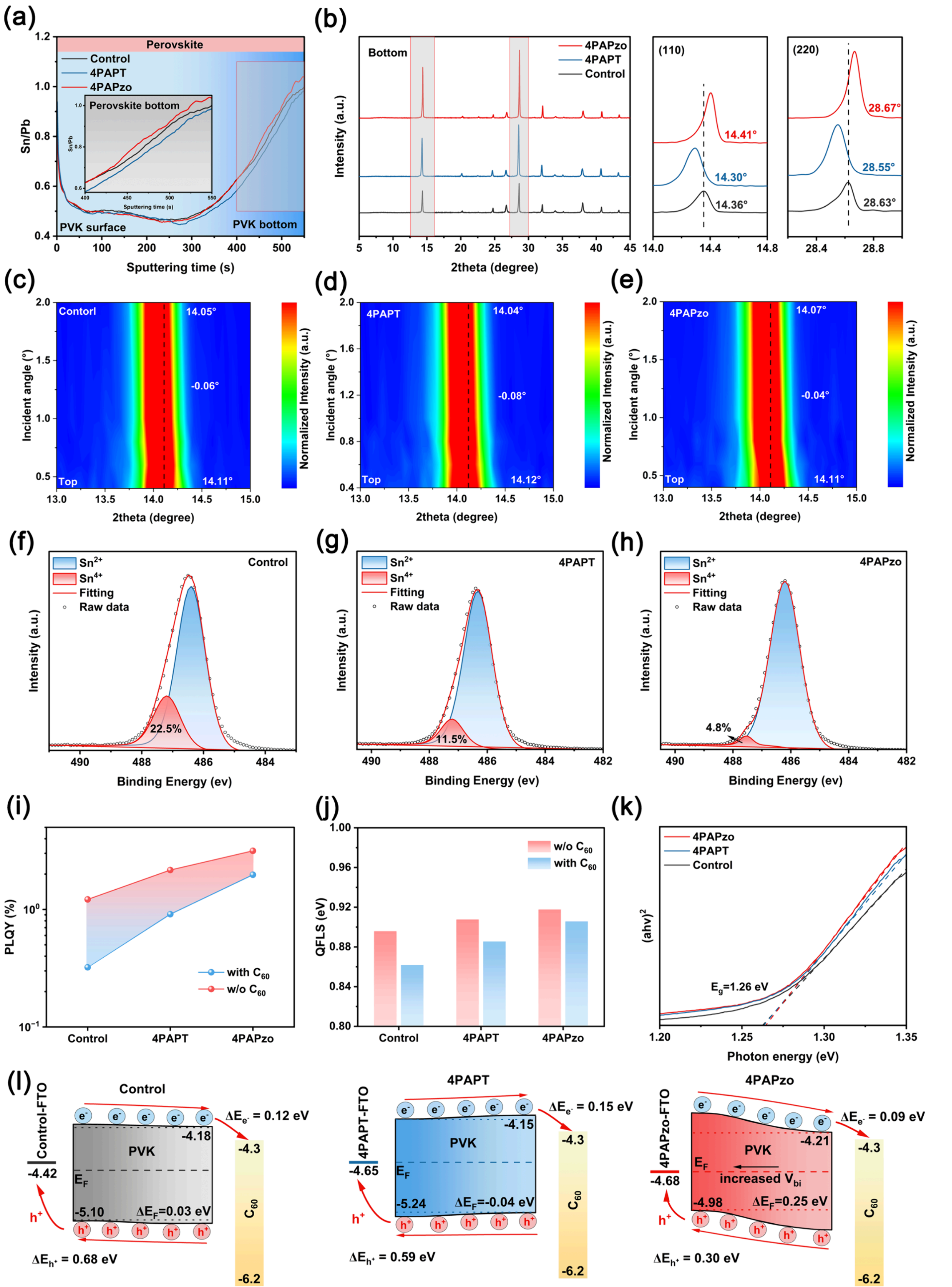

Figure 2. (a) The Sn/Pb ratio in the three perovskite films measured by the TOF-SIMS depth profile. (b) XRD patterns of perovskite films tested from the exposed buried surface side. GIXRD patterns of the buried interface at various incidence angles for (c) Control, (d) 4PAPT-treated, and

Figure 2. continued

(e) 4PAPzo-treated perovskite films. (f, g, h) XPS spectra of Sn 3d peaks for the (f) Control, (g) 4PAPT-treated, and (h) 4PAPzo-treated perovskite films. (i) PLQY and (j) QFLS results for perovskite films with and without C_{60}. (k) Bandgaps of the perovskite films determined from the UV−vis absorption spectra. (l) Schematic illustration of the dark-built-in electric field and hole transport barrier in the three perovskite films.

of 4PAPT and 4PACz. The two O atoms in 4PAPzo have high electronegativity, which increases the dipole moment of the 4PAPzo molecule and favors surface dipole formation. Furthermore, due to the introduction of the additional sulfonyl group, the distribution of the highest occupied molecular orbital (HOMO) of 4PAPzo is slightly expanded compared with that of 4PAPT and 4PACz (Figure S6). In addition, introducing the sulfonyl group enhances the thermal stability of 4PAPzo. As shown by thermogravimetric analysis (TGA) (Figure S7), the temperatures at which 4PAPzo, 4PAPT, and 4PACz undergo 5% weight loss are 334.13 °C, 267.5 °C, and 281 °C,[37] respectively. DFT and experiments indicate that 4PAPzo, designed with the introduction of a sulfonyl group, exhibits stronger surface potential, larger dipole moment, and higher thermal stability compared to 4PAPT and 4PACz.

The sulfonyl group with an orthogonal structure in 4PAPzo creates a steric effect that prevents SAMs from aggregating and jamming during perovskite film formation. As predicted by DFT calculations for p-p stacking of molecules, the binding energies of the dimers formed by 4PAPzo, 4PAPT, and 4PACz are −1.24 eV, −1.30 eV, and −1.44 eV, respectively (Figure S8). The initial configurations of the 4PAPzo and 4PAPT dimers are head-to-head p-p stacking. However, after optimization, 4PAPzo adopts a head-to-tail stacking configuration. This rearrangement occurs primarily because the steric repulsion from the orthogonal sulfonyl group outweighs the p-p interaction, favoring the head-to-tail arrangement and resulting in a more stable intermolecular configuration. In 4PAPT molecules, there is only weak steric hindrance, while the intermolecular p-p stacking interaction is stronger. As a result, the configuration of 4PAPT remains head-to-head stacking after optimization, but its binding energy is much lower than that of 4PACz. In contrast, 4PACz lacks any steric hindrance, resulting in the highest dimer binding energy and a greater tendency to aggregate. Time-of-flight secondary ion mass spectrometry (TOF-SIMS) was used to characterize the distribution of SAMs within the perovskite film (Figure 1c). The results show that the vast majority of 4PAPzo and 4PAPT molecules are well deposited at the bottom interface of the perovskite layer. This is attributed, on one hand, to the steric hindrance in the molecular structures of both compounds, which suppresses their excessive aggregation in the perovskite phase, and, on the other hand, to the phosphonic acid groups in the molecules which further enhance their distribution at the bottom interface through strong bonding interactions with hydroxyl groups on the substrate.[36]

According to HSAB theory, hard acids tend to combine with hard bases, while soft acids tend to combine with soft bases. Since the Lewis basicity of O is stronger than that of S and the Lewis acidity of Sn^{2+} is stronger than that of Pb^{2+}, it was hypothesized that O has a greater tendency to bind with Sn^{2+}, while S has a stronger tendency to bind with Pb^{2+}. To investigate the interaction between the two types of SAMs and Sn^{2+}/Pb^{2+} in the perovskite precursor solution, DFT was used to calculate their binding energies with SnI_2 and PbI_2, respectively. As shown in Figure 1d, the binding energy of 4PAPT with PbI_2 is higher than that with SnI_2, indicating that 4PAPT is more favorable for binding with PbI_2. In contrast, the binding energy of 4PAPzo with SnI_2 is higher than that with PbI_2, indicating that it is more favorable for binding with SnI_2. The VASP-optimized calculation models of 4PAPT and 4PAPzo with SnI_2 and PbI_2, respectively, are shown in Figure S9. To experimentally validate this opposing interaction, Fourier-transform infrared (FTIR) spectroscopy was employed. The C−S stretching vibration of 4PAPT shifted from 1037.22 cm^{-1},[38] 1232.42 cm^{-1}[39] to 1045.23 cm^{-1}, 1230.93 cm^{-1} for the 4PAPT-PbI_2 adduct and 1044.88 cm^{-1},1030.82 cm^{-1} for the 4PAPT-SnI_2 adduct (Figure 1e), while the S=O stretching vibration of 4PAPzo shifted from 1288.34 cm^{-1},[40] 1081.13 cm^{-1}[41] to 1287.49 cm^{-1}, 1079.42 cm^{-1} for the 4PAPzo-PbI_2 adduct and 1282.16 cm^{-1}, 1074.96 cm^{-1} for the 4PAPzo-SnI_2 adduct (Figure 1f). In addition, X-ray photoelectron spectroscopy (XPS) was also employed to analyze the binding energy shifts in the Pb 4f, Sn 3d, and S 2p spectra to further confirm the interactions between 4PAPzo/4PAPT and Pb^{2+}/Sn^{2+}. As shown in Figure 1g-h, compared with the Control sample, the Pb 4f peak shifts to a higher binding energy by 0.15 eV after adding 4PAPT, while the Sn 3d peak shifts to a higher binding energy by 0.10 eV. This also confirms that 4PAPT has a stronger affinity for Pb^{2+}. In contrast, after adding 4PAPzo, the Pb 4f peak shifts to a higher binding energy by 0.20 eV, while the Sn 3d peak shifts to a higher binding energy by 0.30 eV, indicating that 4PAPzo has a stronger affinity for Sn^{2+}. Similarly, compared with the binding energy shift of S in 4PAPT (0.08 eV), the shift of the S peak in 4PAPzo is larger (0.13 eV) (Figure S10). The XPS results further demonstrate that 4PAPzo exhibits a higher binding affinity for both PbI_2 and SnI_2 compared to 4PAPT, making it more effective in passivating the undercoordinated Sn^{2+} and Pb^{2+} in the perovskite film.

Based on DFT calculations and experimental results, we speculate that while 4PAPzo and 4PAPT spontaneously accumulate at the bottom of the perovskite layer, they may alter the concentration ratio of Sn and Pb components at the bottom of the perovskite layer. These differences in the vertical distribution of Sn^{2+} in Sn−Pb perovskite, caused by the doping of SAM molecules, should affect the formation of the built-in electric field.

Based on the analysis and inference from the aforementioned results, it is suggested that the bottom of the perovskite film doped with 4PAPzo may exhibit a higher Sn component compared to the Control due to the strong affinity between 4PAPzo and Sn^{2+}, while the bottom of the perovskite film doped with 4PAPT may show a more pronounced deficiency of the Sn component since 4PAPT preferably binds to Pb^{2+}. Time-of-flight secondary ion mass spectrometry (ToF-SIMS) was used to investigate the distribution of Sn^{2+} and Pb^{2+} in the film, as shown in Figure 2a. Near the bottom of the perovskite films, the Sn/Pb ratio at the bottom of the 4PAPT-treated film is lower than that of the Control, while the Sn/Pb ratio in the 4PAPzo-treated film is higher than that of the Control. This indicates that the bottom of the 4PAPT-treated film contains more Pb perovskite than the Control, whereas the bottom of

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the 4PAPzo-treated film has a higher content of Sn perovskite than the Control.

The perovskite was peeled from the substrate using UV adhesive to characterize the buried interface of perovskite by X-ray diffraction (XRD). The diffraction peaks near 14° and 28° correspond to the (110) and (220) crystal planes of the Sn−Pb perovskite, respectively.[12] As shown in Figure 2b, the (110) and (220) diffraction peaks of the Control films are located at 14.36° and 28.63°, respectively. For the 4PAPT-treated film, the (110) (14.30°) and (220) (28.55°) peaks shift to lower angles by 0.06° and 0.08°, respectively, indicating lattice expansion. In contrast, the (110) (14.41°) and (220) (28.67°) peaks of the 4PAPzo-treated film shift to higher angles by 0.05° and 0.04°, respectively, suggesting lattice contraction. The atomic radius of Sn^{2+} is smaller than that of Pb^{2+}. According to Bragg's Law, an increase in the content of $[SnI_6]^{4-}$ leads to a higher diffraction angle of XRD diffraction peaks, whereas an increase in the content of $[PbI_6]^{4-}$ results in a lower diffraction angle of XRD diffraction peaks.[12,42] Therefore, the XRD results also confirm that the buried interface of the 4PAPzo film contains more Sn components than the Control, while the buried interface of the 4PAPT film contains more Pb components than the Control.

To further investigate the crystal structure of the buried interface of perovskite films, Grazing-incidence X-ray diffraction (GIXRD) was employed for the analysis of the perovskite films.[12] The films were tested at incident angles ranging from 0.4° to 2.0°. A larger incident angle corresponds to a greater detection depth of the film, allowing for closer proximity to the bottom interface of the perovskite film. As shown in Figure 2c-e and S12, as the incident angle increases (scanning from the top to the bottom of the film), the (100) diffraction peak near 14° exhibits different shifting trends. The peak of the Control film shifts to a lower angle by 0.06° (from 14.11° to 14.05°), and the peak of the 4PAPT-treated film shifts to a lower angle by 0.08° (from 14.12° to 14.04°), implying a higher content of Pb perovskite near the buried interface. In contrast, the peak of the 4PAPzo-treated film shifts to a lower angle by only 0.04° (from 14.11° to 14.07°), indicating a higher content of Sn perovskite at the buried interface. Furthermore, from a horizontal comparison of the three films, when the incident angle is 2°, the (100) diffraction peak positions of the Control film, 4PAPT-treated film and 4PAPzo-treated film are 14.05°, 14.04° and 14.07°, respectively. This trend is consistent with the results of XRD characterization of the buried interface, both of which confirm that the buried interface of the 4PAPT-treated film contains a higher content of Pb perovskite than the Control, while the buried interface of the 4PAPzo-treated film has a higher content of Sn perovskite than the Control.

Steady-state photoluminescence (PL) measurements were conducted to further investigate compositional differences between the surface and the buried interface of the perovskite films (Figure S13). In the Control film, the emission peak red-shifted from 988 nm at the top surface to 996 nm at the bottom, a red-shift of 8 nm. For the 4PAPT-treated film, the emission peak red-shifted from 988 nm at the top to 994 nm at the bottom, a red-shift of 6 nm, indicating a higher Pb perovskite content at the bottom compared to the Control.[12] In contrast, for the 4PAPzo-treated film, the emission peak red-shifted from 986 nm at the top to 997 nm at the bottom, a red-shift of 11 nm, suggesting a greater content of Sn perovskite content at the bottom.[12] Furthermore, due to the interaction between the SAM and perovskite components, the 4PAPzo-treated film exhibited the lowest defect density and the highest PL emission intensity, followed by the 4PAPT-treated film and the Control film.

To investigate the effects of both improved Sn distribution and the application of SAMs on reducing p-doping at the upper interface caused by Sn^{4+}, XPS measurements were conducted on the surfaces of the perovskite films (Figure 2f-h). Due to the regulation of Sn components and the strong interaction between SAMs and Sn^{2+}, the results indicated that the Sn^{2+} content decreased from 22.5% in the Control film to 11.5% and 4.8% in the 4PAPT-treated and 4PAPzo-treated films, respectively. Therefore, it can be inferred that the application of SAMs enhances the n-type characteristics of the top interface, thereby increasing the top-interface built-in electric field and promoting electron transport. To verify this hypothesis, UPS measurements were conducted on the top surface of the perovskite films. The results show that the distance between the HOMO and the E_F for the Control, 4PAPT-treated, and 4PAPzo-treated films is 0.69, 0.71, and 0.75 eV, respectively (Figure S14). This suggests that the 4PAPzo-treated perovskite film exhibits a stronger n-type characteristic, which contributes to the enhancement of the top-interface built-in electric field. Furthermore, to study the effect of Sn distribution of buried interface on the energy levels at the bottom interface of the perovskite films and the bottom-interface built-in electric field, UPS measurements were also performed on the bottom interface (Figure S15). The results reveal that for the Control, 4PAPT-treated, and 4PAPzo-treated films, the HOMO energy levels are −5.10 eV, −5.24 eV, and −4.98 eV, respectively, with corresponding distances between the HOMO and E_F of 0.66, 0.75, and 0.50 eV. The increase in Sn components at the bottom interface in the 4PAPzo-treated films enhances the VBM, leading to a stronger p-type characteristic at the bottom interface and facilitating the enhancement of the bottom-interface built-in electric field, thereby promoting hole transport.

To quantitatively assess the enhancement of the built-in electric field at the interface, we conducted photoluminescence quantum yield (PLQY) measurements and quasi-Fermi level splitting (QFLS) analyses (Figure 2i-j). For perovskite films without C_{60}, the PLQY increased from 1.22% in the Control sample to 2.16% in the 4PAPT sample and 3.15% in the 4PAPzo sample. However, with the addition of C_{60}, the PLQY values for the Control and 4PAPT samples decreased to 0.32% and 0.91%, respectively, indicating significant nonradiative recombination due to insufficient built-in electric field and the presence of defects. In contrast, the PLQY for the 4PAPzo sample was measured at 1.97%, suggesting that enhanced built-in electric field significantly facilitated charge carrier transport. Consequently, the QFLS values with C_{60} were calculated to increase from 0.862 eV in the Control sample to 0.885 and 0.906 eV in the 4PAPT and 4PAPzo samples, respectively. These results indicate that enhancing the built-in electric field can suppress QFLS losses, which is beneficial for improving the V_{OC} of the devices.

Additionally, ultraviolet−visible (UV−vis) absorption spectroscopy was employed (Figure 2k and S.16) to determine the band gaps of the three perovskite films and to construct their energy band diagrams (Figure 2l). In the Control film, the dark built-in electric field between the top and bottom is only 0.03 eV. In the 4PAPT-treated film, the increased Pb perovskite content at the bottom causes its VBM to decrease from −5.10 eV (Control) to −5.24 eV. As a result, the dark-built-in electric

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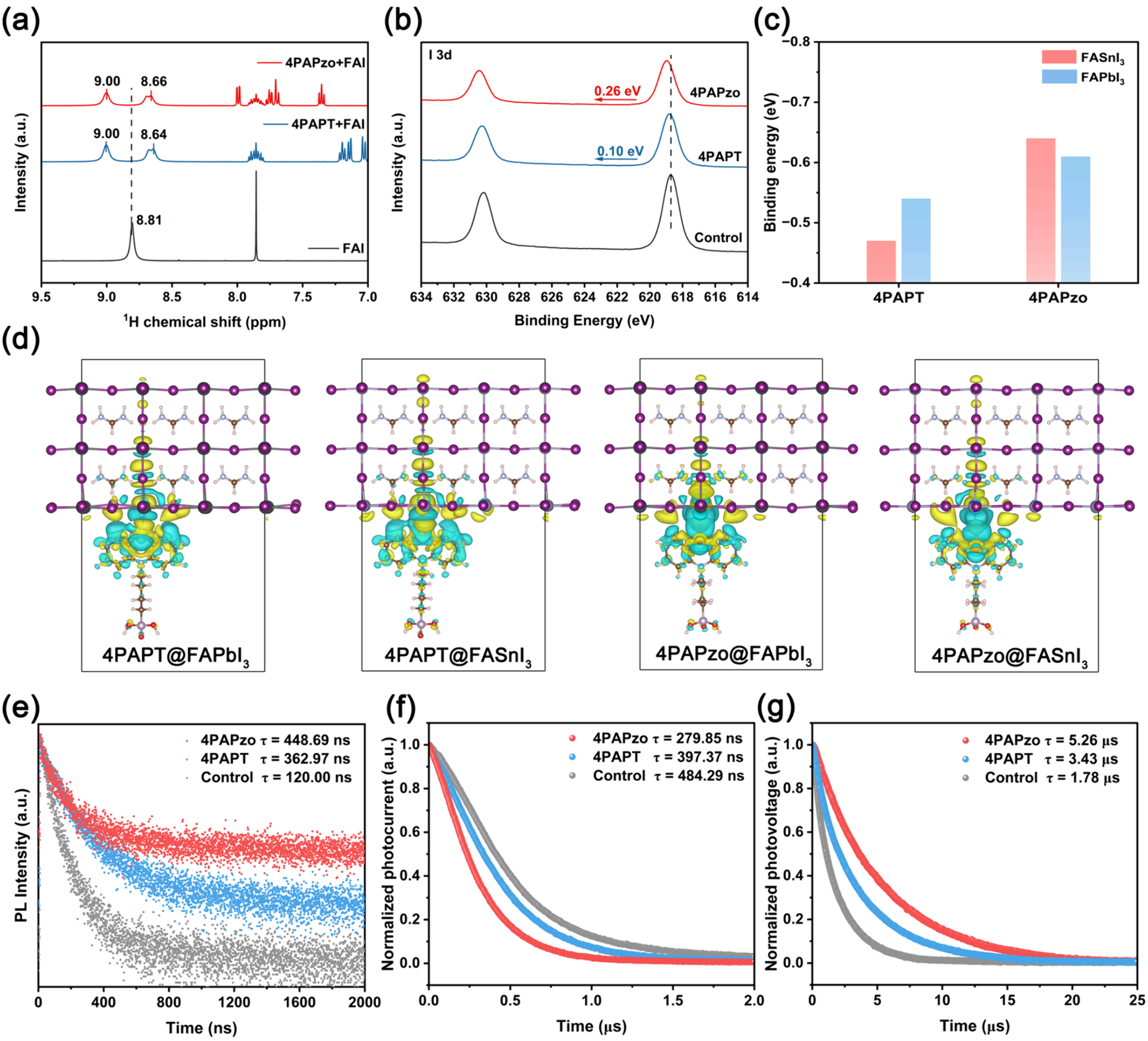

Figure 3. (a) ^{1}H NMR spectra of FAI and mixtures. (b) XPS spectra of I 3d at the perovskite film. (c) The binding energy between SAMs and $FAPbI_3$/$FASnI_3$. (d) Charge density differences of 4PAPT and 4PAPzo on the buried interface of $FAPbI_3$ and $FASnI_3$. (The yellow region indicates electron accumulation, while the cyan region indicates electron depletion). (e) TRPL results of perovskite films. (f) TPC decay curves of Sn−Pb PSCs. (g) TPV decay curves of Sn−Pb PSCs.

field from the top to the bottom changes to −0.04 eV, and the direction of this dark-built-in electric field is opposite to the direction of carrier transport, which is unfavorable for the directional transport of carriers. The charge extraction force of the first two samples mainly originates from the energy offset at the interfaces between the electron-transport-layer (ETL) and the HTL. In contrast, the increased Sn perovskite content at the bottom of the 4PAPzo-treated film raises the bottom VBM from −5.10 eV (Control) to −4.98 eV, establishing a 0.25 eV dark-built-in electric field within the perovskite layer. This increased electric field helps to promote the separation and directional transport of holes and electrons within the perovskite layer when there is no HTL. Kelvin probe force microscopy (KPFM) measurements were performed on the top and bottom surfaces of the perovskite films (Figure S17−18), and the variation trend of the work function (W_F) was consistent with that observed in the UPS measurements. Furthermore, it can be observed that the contact potential difference (CPD) at the buried interface of the 4PAPzo-treated film exhibits a more uniform distribution compared to that of the Control and 4PAPT-treated film (Figure S18). This phenomenon may be attributed to reduced nonradiative recombination and the passivation effect of 4PAPzo at the buried interface of the perovskite film.

To determine the hole transport barrier between the perovskite and the bottom interface, UPS measurements were performed on the fluorine-doped tin oxide (FTO) substrate after peeling off the perovskite layer with UV adhesive (Figure S19). The work function W_F of the FTO substrate contaminated by the Control perovskite was −4.42 eV. For substrates contaminated by the perovskites with 4PAPT and 4PAPzo added, the surface W_F shifted downward

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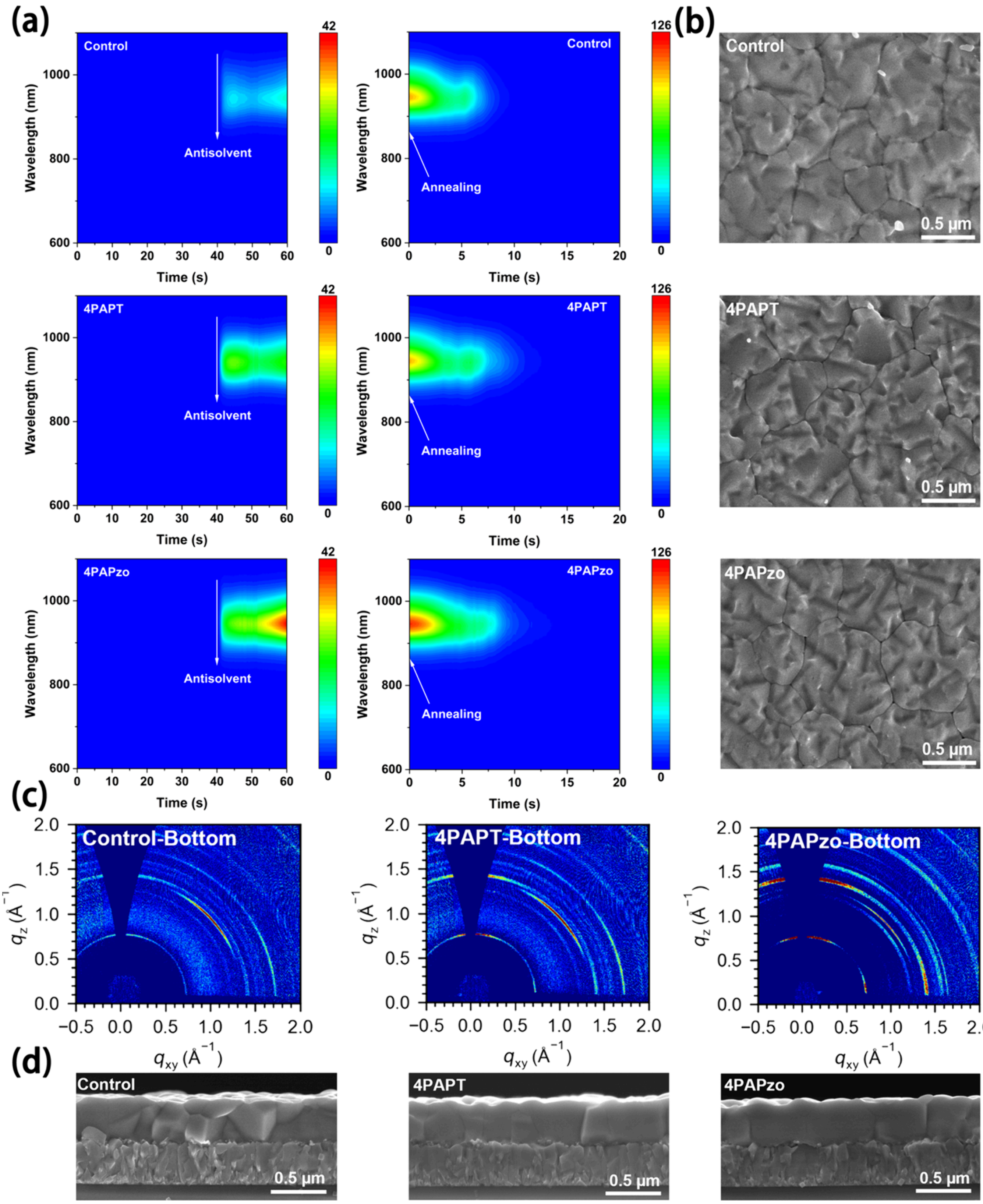

Figure 4. (a) *In-situ* PL spectra during spin-coating and annealing stages of perovskite films. (b) Top-view SEM images of the buried surface of perovskite films. (c) GIWAXS patterns of the buried surface of perovskite films. (d) Cross-section SEM images of the perovskite films.

to −4.65 eV and −4.68 eV, respectively. Correspondingly, the hole transport barrier of the Control device is 0.68 eV. Although the VBM at the buried interface of the 4PAPT-treated perovskite is slightly lower than that of the Control device, the hole transport barrier is reduced to 0.59 eV due to the deepening of the surface W_F of the FTO substrate facilitated by 4PAPT. In contrast, the 4PAPzo-treated device exhibits the lowest hole transport barrier, only 0.30 eV, due to

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the elevation of the VBM at the buried interface and the deepening of the W_F of the FTO substrate caused by 4PAPzo. In addition, the electron transport barriers[43] for the Control, 4PAPT, and 4PAPzo samples are 0.12, 0.15, and 0.09 eV, respectively. The 4PAPzo sample exhibits the lowest hole transport barrier and electron transport barrier.

The results of the aforementioned characterization tests confirm our hypothesis: the introduction of 4PAPzo selectively induces the accumulation of Sn components at the perovskite buried interface. This phenomenon effectively alleviates the common issue of Sn deficiency at the bottom of the Sn−Pb perovskite layer, reduces p-doping at the top interface, and enhances its n-type character, thereby strengthening the built-in electric field at the top interface. Furthermore, the increase in the Sn component at the bottom interface of the perovskite promotes the upward shift of its VBM, making the bottom interface more p-type and strengthening the built-in electric field at the bottom interface. This not only establishes a stable built-in electric field within the perovskite layer, facilitating the directional separation and transport of holes and electrons but also reduces the hole transport barrier at the bottom interface and the electron transport barrier at the top interface.

To investigate the bonding strength of 4PAPT and 4PAPzo with iodide ions (I^-), we separately dissolved the two SAMs with formamidinium iodide (FAI) in deuterated dimethyl sulfoxide (DMSO-d_6) and performed nuclear magnetic resonance spectroscopy (^{1}H NMR). As shown in Figure 3a, the N−H proton of pristine FAI exhibits a chemical shift at 8.81 ppm. After mixing FAI with 4PAPzo, the peak splits into two major peaks: one shifts downfield to 9.00 ppm and the other shifts upfield to 8.66 ppm. A similar shift pattern is observed upon mixing with 4PAPT, demonstrating that both 4PAPT and 4PAPzo can form hydrogen bonds with FAI. Furthermore, XPS measurements were also performed on the films to further verify the interfacial interactions between I^- and the SAMs. As shown in Figure 3b, the XPS peak of I^- in the 4PAPT film shifts to a higher binding energy by 0.10 eV, while the peak in the 4PAPzo film shifts to a higher binding energy by 0.26 eV. This indicates that 4PAPzo has a stronger passivation ability for I^- defects. DFT was used to determine the binding energies of 4PAPT and 4PAPzo with $FASnI_3$ and $FAPbI_3$, respectively, to verify the passivation ability of the two SAMs toward the uncoordinated B-site atoms on the surface (Figure 3c and Figure S21). The binding energies of 4PAPT with undercoordinated Sn^{2+} and Pb^{2+} on the surfaces of $FASnI_3$ and $FAPbI_3$ are 0.47 and 0.54 eV, respectively. In contrast, 4PAPzo exhibits higher binding energies of 0.64 and 0.61 eV with undercoordinated Sn^{2+} and Pb^{2+} on the respective surfaces, demonstrating its superior passivation capability toward undercoordinated B-site ions.

To investigate the effect of the interaction between SAMs and perovskite on hole transport, the charge density difference (CDD) was calculated (Figure 3d), and the corresponding charge density difference along the z-axis perpendicular to the buried surface of the perovskite was derived (Figure S22). The CDD plots show that both SAMs, 4PAPT and 4PAPzo, induce a redistribution of charges at the buried interface of the perovskites. The negative charges (cyan regions) are mainly distributed on the SAMs side, while the positive charges (yellow regions) are mainly concentrated on the perovskite side. This indicates that holes are energetically more likely to transfer from the perovskite to the SAMs.[44] Therefore, the coordination interaction between the SAMs and the buried interface of the perovskites further promotes the transport and extraction of holes. In addition, the results of transient reflectance (TR) measurements on the hole extraction dynamics at the bottom interface of the three samples show that the 4PAPzo-modified sample exhibits higher values of both the carrier diffusion coefficient (D) and the surface extraction velocity (S) (Figure S23), which further confirms its role in promoting hole extraction.[45]

To further investigate differences in hole transport performance, time-resolved photoluminescence (TRPL) measurements were conducted through the glass substrate side, and the PL decay curves were fitted using a biexponential rate law model (Figure 3e and Table S1). For the Control film, the carrier lifetimes are 140.28 ns (τ_1) and 93.46 ns (τ_2). For the 4PAPT-treated film, the carrier lifetimes are 128.86 ns (τ_1) and 440.35 ns (τ_2). For the 4PAPzo-treated film, the carrier lifetimes are 119.57 ns (τ_1) and 553.55 ns (τ_2). The average carrier lifetimes for the Control, 4PAPT-treated, and 4PAPzo-treated films are 120.00, 362.97, and 448.69 ns, respectively. The longest average carrier lifetime observed in the 4PAPzo-treated film further indicates that the improvement in Sn component distribution and the suppression of defects have reduced the nonradiative recombination. In addition, carrier transport and recombination characteristics were further investigated by measuring transient photocurrent decay (TPC) (Figure 3f) and transient photovoltage decay (TPV) (Figure 3g). The TPC decay times for the Control, 4PAPT, and 4PAPzo PSCs are 484.29, 397.37, and 279.85 ns, respectively. The shortest TPC decay time observed for the 4PAPzo device indicates improved charge extraction.[46] The TPV decay times for the Control, 4PAPT, and 4PAPzo PSCs are 1.78 μs, 3.43 μs, and 5.26 μs, respectively. The significantly increased photovoltage lifetime indicates reduced surface recombination and improved film quality in the devices treated with 4PAPzo.[46]

These results indicate that 4PAPzo can passivate I^- defects by forming hydrogen bonds with FAI. DFT calculations also confirm that 4PAPzo exhibits higher binding energy to uncoordinated Sn^{2+} and Pb^{2+} in $FASnI_3$ and $FAPbI_3$, demonstrating superior passivation capability for B-site defects. Additionally, 4PAPzo induces charge redistribution that favors hole transport. Devices treated with 4PAPzo achieved the fastest charge extraction rates, longest carrier lifetimes, and significantly suppressed nonradiative recombination.

To investigate the effects of 4PAPT and 4PAPzo on the crystallization kinetics of Sn−Pb perovskite films, *in situ* photoluminescence (PL) spectra were collected during the spin-coating and annealing stages (Figure 4a). Before the antisolvent was added, none of the three films exhibited a photoluminescence (PL) signal, corresponding to the precrystallization stage of the perovskite. When the antisolvent was dropped at the 40th second of spin-coating process, the PL signal intensity increased rapidly, indicating the onset of rapid nucleation after the removal of the polar solvent. After continuing the spin-coating for an additional 10 s following the drop of the antisolvent and then monitoring for another 10 s after the spin-coating was completed, the PL intensity of the films reached its peak. Among the three films, the PL intensity of the 4PAPzo-treated film was the highest. The peak intensity at 950 nm was selected to create a graph relative to time (Figure S24), which shows that the PL intensity of the 4PAPzo film is the highest. This is mainly because the enrichment of 4PAPzo at the bottom provides more nucleation sites, resulting

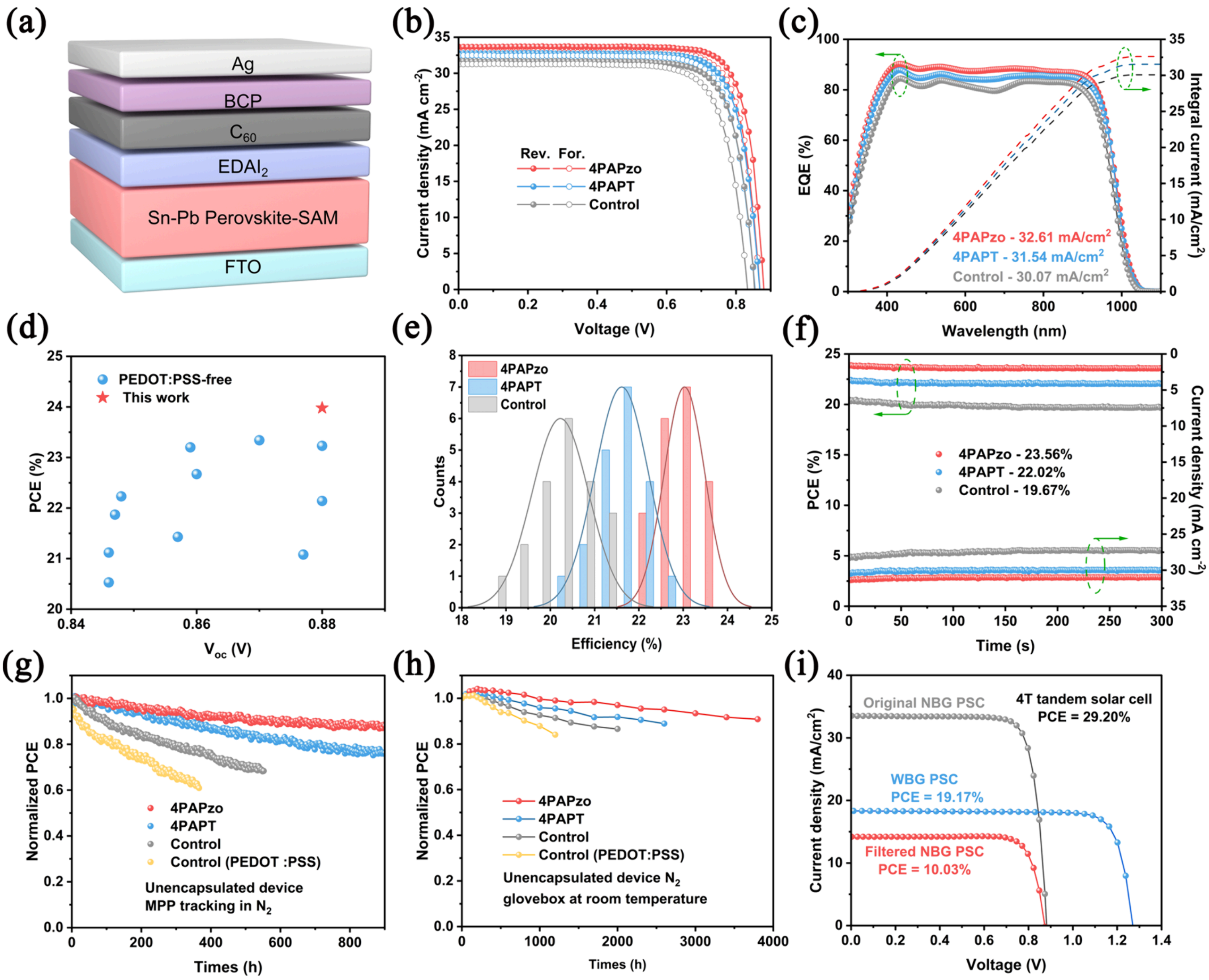

Figure 5. (a) Device architecture of Sn–Pb PSC. (b) J–V curves of the champion Control, 4PAPT-modified, and 4PAPzo-modified devices. (c) EQE spectra and integrated J_{SC} of Sn–Pb devices. (d) The PCE and V_{OC} statistics of our champion Sn–Pb PSC and representative previous studies about PEDOT:PSS-free Sn–Pb PSCs. (e) Distribution histograms of the PCE for Control, 4PAPT, and 4PAPzo devices. (f) SPO efficiencies of the Control, 4PAPT, and 4PAPzo devices. (g) MPPT results of unencapsulated devices measured under simulated one-sun illumination in nitrogen. (h) Shelf-storage stability of unencapsulated Sn–Pb devices in nitrogen. (i) J–V characteristics curves of semitransparent WBG device and filtered NBG device.

in the most rapid nucleation.[47] During the annealing stage (Figure 4a and S25), the PL intensity gradually decreased due to thermally induced defects and nonradiative recombination. Notably, the 4PAPzo sample exhibited the slowest decline in PL intensity, suggesting that the interaction between 4PAPzo and SnI_2 significantly slowed the perovskite crystallization process and promoted the growth of larger grains.[47]

To investigate the impact of SAMs on film quality, the morphology of the buried surface, top surface, and cross-section of the perovskite films was characterized using scanning electron microscopy (SEM). As shown in the buried interfaces of the three films in Figure 4b and S26, and their top surfaces in Figure S27–S28, the Control film exhibits inhomogeneous and relatively small grains due to its uncontrolled and rapid crystallization process. Due to a certain crystallization delay, the grain size of the 4PAPT-treated film slightly increases, while the grains in the 4PAPzo-treated film are uniform and the largest. Furthermore, grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were conducted on the buried surfaces of the perovskite films at an incident angle of 0.3° (Figure 4c). Both the top and bottom surfaces of the three films exhibit a preferential orientation along the (100) direction. For both the top and buried surfaces, the 4PAPzo-treated film exhibits the highest diffraction intensity along the (100) direction, followed by the 4PAPT-treated film and the Control film, indicating that 4PAPzo effectively enhances the crystallographic orientation of the Sn–Pb perovskite (Figure 4c and Figure S29). The cross-sectional SEM images also show consistent results (Figure 4d and Figure S30). In the Control film, longitudinal grain boundaries, which may hinder carrier transport, are observed, along with a disordered and smaller grain size distribution. In the 4PAPT-treated film, the longitudinal grain boundaries are significantly reduced, and there is a clear vertical orientation. The 4PAPzo-treated film demonstrates the most regular vertical orientation and the largest grain size. Furthermore, atom force microscopy (AFM) measurements revealed that the surface roughness root-mean-square (RMS) values of the Control, 4PAPT-treated, and

4PAPzo-treated films are 33.5, 31.6, and 25.1 nm, respectively (Figure S31). The 4PAPzo-treated film exhibits the lowest RMS, which helps improve the interface contact between the perovskite layer and the electron transport layer (ETL).[48]

The above results indicate that introducing 4PAPzo effectively regulates the crystallization dynamics of Sn−Pb perovskites, promoting the formation of higher-quality perovskite films with stronger vertical orientation and larger, more uniform, smoother morphology, thus providing a solid foundation for further improvement of device performance.

An inverted single-junction Sn−Pb perovskite solar cell (PSC) with the configuration, FTO/perovskite/$EDAI_2$/C_{60}/bathocuproine (BCP)/Ag, was fabricated (Figure 5a), and the current density−voltage ($J-V$) curves of the device were measured under simulated AM 1.5 G illumination. The $J-V$ curves of devices doped with different concentrations of 4PAPT and 4PAPzo are shown in Figure S32. It can be observed that the optimal concentration of both SAMs is 0.5 mg/mL. As shown in Figure 5b, the best PCE of the control device (without PEDOT:PSS or SAM) was 21.19%, with an open-circuit voltage (V_{OC}) of 0.853 V, a short-circuit current density (J_{SC}) of 31.68 mA cm^{-2}, and a fill factor (FF) of 78.37%. After the introduction of 4PAPT, the PCE of the Sn−Pb PSC increased to 22.51% (V_{OC} = 0.868 V, J_{SC} = 32.75 mA cm^{-2}, FF = 79.18%). The device treated with 4PAPzo achieved a champion PCE of 23.98% (V_{OC} = 0.880 V, J_{SC} = 33.66 mA cm^{-2}, and FF = 80.98%). The champion device modified with 4PAPzo exhibited an integrated J_{SC} of 32.61 mA cm^{-2} in the external quantum efficiency (EQE) measurement, which is in good agreement with the $J-V$ results (Figure 5c). Notably, the champion efficiency of the device modified via 4PAPzo codeposition ranks among the highest reported values for Sn−Pb PSCs without PEDOT:PSS (Figure 5d and Table S2). The PCE histograms from 20 individual devices under each of the three conditions confirm the reproducibility of the efficiency improvement (Figure 5e). The average PCEs of devices modified with 4PAPT and 4PAPzo increased from 20.22% to 21.61% and 23.03%, respectively (Figure 5e). The enhancement in photovoltaic performance is primarily attributed to the establishment of an enlarged built-in electric field and the regulation of crystallization quality.

Dark current−voltage ($I-V$) measurements were used to evaluate the effect of the additive on carrier dynamics in the device (Figure S33).[49] The results show that the leakage current in the device modified with 4PAPzo is significantly reduced, indicating a decrease in device defects and carrier recombination. Space-charge-limited current (SCLC) measurements were conducted to quantitatively investigate the trap density (N_t) in the perovskite film using hole-only devices.[50] The trap-filled limit voltage (V_{TEF}) and N_t values of the perovskite film were extracted from the $I-V$ results presented in Figure S34, with both parameters summarized in Table S3. The V_{TEF} values for the Control, 4PAPT, and 4PAPzo devices are 0.66, 0.55, and 0.35 V, respectively. The corresponding N_t values for the perovskite films are 7.37×10^{15} cm^{-3}, 6.14×10^{15} cm^{-3}, and 3.91×10^{15} cm^{-3}, respectively. This is mainly attributed to the fact that SAM molecules inhibit iodide ion migration and provide a passivation effect on undercoordinated Sn^{2+} and Pb^{2+}, thereby reducing the defect density in the device.

The stability of the devices was further investigated. The stabilized power output (SPO) at the maximum power point (MPP) for the Control, 4PAPT-modified, and 4PAPzo-modified devices was 19.67%, 22.02%, and 23.56%, respectively (Figure 5f). The operational stability was evaluated under simulated AM1.5 light exposure in a nitrogen glovebox at the MPP (Figure 5g). The control device with PEDOT:PSS retained 60.89% of its initial PCE after 366 h. The control device without HTL retained 68.34% of its initial PCE after 552 h. The 4PAPT-modified and 4PAPzo-modified devices retained 76.22% and 88.00% of their initial PCE after 900 h, respectively. In addition, the shelf-storage stability was tested without encapsulation in nitrogen and dark (Figure 5h). The Control device with PEDOT:PSS retained 84.04% of its initial PCE after 1200 h. The Control device without HTL retained 86.56% of its initial PCE after 2000 h. The 4PAPT-modified device retained 88.90% of its initial PCE after 2600 h. The 4PAPzo-modified device preserved 90.83% of its initial PCE after 3800 h. For thermal stability, unencapsulated Sn−Pb devices were aged at 85 °C in a nitrogen atmosphere. As shown in Figure S35, the Control device based on PEDOT:PSS retained only 60.17% of its initial efficiency after 250 h of aging. The Control device without HTL exhibited slightly improved stability, retaining 66.98% of its initial efficiency after 360 h. In contrast, the device optimized with 4PAPT exhibited superior thermal stability, retaining 76.45% of its initial efficiency after 430 h of aging. Notably, the device modified with 4PAPzo displayed the best thermal stability among all groups, retaining 85.19% of its initial efficiency after 500 h of continuous aging. These stability test results highlight the role of 4PAPzo in stabilizing the built-in electric field and enhancing device stability.

Based on the optimized Sn−Pb NBG PSC, a 4-T all-perovskite tandem solar cell was fabricated. The schematic of the tandem device is shown in Figure S36. The semitransparent WBG device with a bandgap of 1.77 eV served as the top cell, while the NBG device served as the bottom cell. The PCE of the single WBG PSC device is 20.94% (Figure S37 and Table S4). The $J-V$ curve of the 4-T device is shown in Figure 5i. The semitransparent WBG PSC achieved a PCE of 19.17%, while the filtered NBG PSC achieved a PCE of 10.03%. A detailed summary of the photovoltaic parameters is provided in Table S5. The combined WBG and NBG cells produced a total PCE of 29.20% for the 4-T all-perovskite tandem solar cell, which is among the highest values reported for 4-T all-perovskite TSCs (Table S6). The EQE spectra of the two cells are shown in Figure S38, with integrated J_{SC} values of 17.82 mA/cm^2 for the WBG cell and 13.98 mA/cm^2 for the NBG cell, which are in excellent agreement with the J_{SC} values obtained from the $J-V$ measurements.

In summary, this work addresses the issue of unstable built-in electric fields at interfaces in Sn−Pb perovskite solar cells (PSCs) by introducing a new self-assembled monolayer molecule, 4PAPzo, which contains a sulfonyl group (O=S=O), into the precursor solution for the codeposition process. For comparison, another SAM, 4PAPT, containing a sulfur group, is also demonstrated. The oxygen atoms with strong Lewis basicity in the sulfonyl group of 4PAPzo preferentially coordinate with Sn^{2+} ions, which have strong Lewis acidity, while the sulfur atoms with weak Lewis basicity in 4PAPT preferentially coordinate with Pb^{2+} ions, which have weak Lewis acidity. Therefore, during the codeposition process, 4PAPzo binds more Sn^{2+} and deposits at the bottom interface, alleviating the weakening of the bottom-interface built-in electric field caused by Sn accumulation at the top interface. This component regulation also increases the VBM at the bottom interface, providing favorable conditions for the formation of an enlarged bottom-interface built-in electric field. In addition, removing PEDOT:PSS fundamentally eliminates the chemical reaction in which PSS oxidizes Sn^{2+} to Sn^{4+}. Meanwhile, 4PAPzo acts as an antioxidant, providing electron cloud density to Sn^{2+} through the Lewis basic oxygen atoms in its sulfonyl group, thereby inhibiting the oxidation of Sn^{2+}. XPS results show that the content of Sn^{4+} at the top interface is significantly reduced, decreasing harmful p-doping at the top interface. UPS results further confirm that the energy band at the top interface is more n-type, which further enhances the strength and stability of the top-interface built-in electric field. These results collectively contribute to the formation of a stable 0.25 eV dark-state built-in electric field in Sn−Pb devices without PEDOT:PSS. The enhanced built-in electric field also enables the device to achieve a V_{OC} of 0.880 V. As a result, the PEDOT:PSS-free Sn−Pb PSC fabricated using this codeposition strategy achieved a champion PCE of 23.98%. It also demonstrated excellent stability, retaining 93.83% of its initial efficiency after 3800 h of storage in a N_2 atmosphere and 88% after 900 h of MPP tracking. Furthermore, the 4-T all-perovskite tandem solar cell, fabricated with an NBG PSC as the bottom subcell using this method, achieved a PCE of 29.20%. This work provides a new approach to enhancing the performance of Sn−Pb PSCs and tandem solar cells without PEDOT:PSS or HTL.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsenergylett.6c00210.

> Experimental section includes materials, preparation of precursor solutions, fabrication of solar cells, characterization, calculation methods for binding energy and adsorption energy, SEM images, ^{1}H NMR spectra, FTIR spectra, ToF-SIMS images, GIXRD patterns, XRD patterns, UPS spectra, UV−vis absorption spectra, SCLC plots, device parameters, XPS spectra, SPO tests, EQE spectra, DFT calculation results and parameters, as well as the photovoltaic parameters of PSCs (PDF)

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Notes

The authors declare no competing financial interest.

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